

RouteScope: Auditing Training-Free First-Tool Routing for Multimodal Agents

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Abstract

Multimodal agents increasingly combine language reasoning with external perception tools, but their first action is often chosen by habit rather than by evidence: answering directly preserves the original visual context, while OCR, visual-question answering, zoom, and caption-style transformations can reveal useful information or introduce brittle intermediate summaries. This paper studies that first-action decision through RouteScope, a training-free router that selects the initial evidence source before final answer generation while preventing gold answers, evaluator feedback, benchmark-family labels, and task identifiers from influencing the route. We compare RouteScope with direct answering, fixed-tool controls, literature-inspired tool-use controls, and a caption-style visual control on 60 selected tasks from MMMU/MMMU-Pro, OCRBench/OCRBench variants, and MathVista, yielding a 480-row method-task design with 479 scored rows under a shared hosted-backend budget. The result is negative but diagnostic: RouteScope reaches 31.7% aggregate accuracy, below Direct Answer at 33.3%, while the strongest caption-style control reaches 41.7%; at the family level, zoom leads MathVista, caption-style evidence leads MMMU, and OCRBench remains low for all methods. These results show that first-tool choice is measurable and consequential, but that the evaluated training-free policy is not yet a reliable selector; the next causal question is whether its losses come from choosing the wrong evidence source or from failing to use routed evidence during answer generation.

1 Introduction

Multimodal agent benchmarks increasingly evaluate systems that can interleave language reasoning with external perception and action modules. Tool-using language agents such as ReAct and Toolformer motivate the idea that choosing an external

action can change the answer trajectory rather than merely decorate it (Yao et al., 2022; Schick et al., 2023). Visual question answering and OCR benchmarks make this issue concrete because they score whether visual evidence is actually recovered, not merely whether a tool was invoked (Antol et al., 2015; Liu et al., 2023). Modular multimodal systems such as ViperGPT, MM-REACT, HuggingGPT, and Chameleon show that expert tools can be composed into answer pipelines, but they also leave open when a tool should be the first evidence source for a new row (Suris et al., 2023; Yang et al., 2023; Shen et al., 2023; Lu et al., 2023a). Broad multimodal reasoning benchmarks such as MMMU, MathVista, and VisualWebArena add another pressure: different families reward different visual transformations, so a single fixed first tool is unlikely to be uniformly optimal (Yue et al., 2023; Lu et al., 2023b; Koh et al., 2024).

This paper asks a narrower question than open-ended agent autonomy: before answering a visual-language task, can a training-free policy select the first branch that is most useful for that row? The question matters because the first branch determines which evidence is exposed to the final answerer and which irrelevant transformations may be introduced. A direct-answer branch preserves the model’s original visual context; OCR can reveal text; VQA and zoom can emphasize visual details; captioning or set-of-mark style controls can summarize visual structure. A useful router should avoid treating these tools as universally good.

We evaluate this question using public multimodal benchmark families. The executed family mix is MMMU/MMMU-Pro, OCRBench/OCRBench variants, and MathVista, with 20 selected tasks per family and eight method conditions per task. VisualWebArena is not included in the reported evidence, and MathVista should be read as a visual-math replacement axis rather than equivalent web-interaction evidence. The evalua-

tion contains 480 expected rows, 479 scored rows, and one reported residual MMMU null. This supports a diagnostic paper, not a final positive-result claim.

Our main result is deliberately conservative. RouteScope scores 31.7% aggregate accuracy, below Direct Answer at 33.3%; VWA SoM Caption is the aggregate leader at 41.7%. Per-family rankings nevertheless shift: MathVista: Always Zoom (65.0 accuracy); MMMU: VWA SoM Caption (55.0 accuracy); OCRBench: Always VQA (10.0 accuracy). We therefore frame RouteScope as an audit of first-tool utility and failure modes rather than as a demonstrated aggregate improvement.

2 Related Work

Tool-use language agents. ReAct studies interleaved reasoning and acting, while Toolformer asks language models to learn when external APIs are useful (Yao et al., 2022; Schick et al., 2023). HuggingGPT and Chameleon broaden this pattern into controller-style systems that delegate subtasks to specialist models or tools (Shen et al., 2023; Lu et al., 2023a). RouteScope differs in scope: it does not train a new tool-use model or claim broad autonomous planning, but instead evaluates a first-action routing policy against direct and fixed-tool controls.

Multimodal expert-tool systems. VQA establishes answer-scored visual question answering as a core evaluation format (Antol et al., 2015). ViperGPT and MM-REACT motivate multimodal reasoning pipelines that invoke external visual or programmatic modules (Suris et al., 2023; Yang et al., 2023). These systems make the routing problem concrete: OCR, VQA, zoom, captioning, and code are not interchangeable. The present study measures that heterogeneity with row-level answer scores rather than qualitative demonstrations.

Benchmarks for multimodal agents. OCR-Bench targets hidden OCR capabilities in multimodal models, MMMU tests expert multimodal reasoning, MathVista emphasizes mathematical and visual reasoning, and VisualWebArena evaluates realistic visual web tasks (Liu et al., 2023; Yue et al., 2023; Koh et al., 2024). Our executed matrix covers OCRBench, MMMU, and MathVista. VisualWebArena remains a relevant benchmark target, but it is not part of the reported experiments.

3 Method

RouteScope is a training-free first-action router. For each row, the controller receives only task-facing inputs: the question, answer choices when present, image or observation content made available by the benchmark interface, and outputs from tools that are legitimately called within the same episode. The controller does not receive gold answers, evaluator outputs, source split identifiers, benchmark-family labels, subject metadata, or task identifiers as routing signals.

The evaluated action set has one direct branch and multiple tool-mediated branches. Direct Answer produces an answer from the original task context. Always OCR, Always VQA, and Always Zoom are fixed-tool controls. MM-REACT Style and ViperGPT Style are literature-inspired controls that emulate multimodal tool use patterns. VWA SoM Caption is a captioning/set-of-mark style visual control evaluated on the selected families; its name is a method condition and does not imply VisualWebArena family execution. RouteScope selects a first branch using the same row contract and then answers from the resulting evidence.

The design goal is not to maximize a hidden oracle. A row is counted only when the answer is produced and scored by the benchmark-facing scorer. Null rows are not converted into wins or losses by hand; they remain null, enter row-count accounting, and are excluded only from score means. This matters for the current evidence because one Always-OCR MMMU-Pro row has no scored answer.

The router is intentionally evaluated as a first-action policy rather than a full planner. A full planner could call several tools, inspect intermediate failures, and revise its plan. That setting is important for deployment, but it makes tool utility hard to attribute because the final answer may depend on a long sequence of actions. The first-action protocol asks a more auditable question: given the same row and budget family, which initial evidence source should be exposed before the answer is produced? This design lets the matrix compare direct answering, fixed tools, literature-style controls, and RouteScope with the same task denominator.

RouteScope’s input contract is also part of the method. The policy may inspect the natural-language question, answer options, and image-facing observation available to the answerer. It may not inspect the benchmark family, split name, task identifier, subject tag, gold answer, evaluator feed-

RouteScope: Training-Free First-Tool Routing

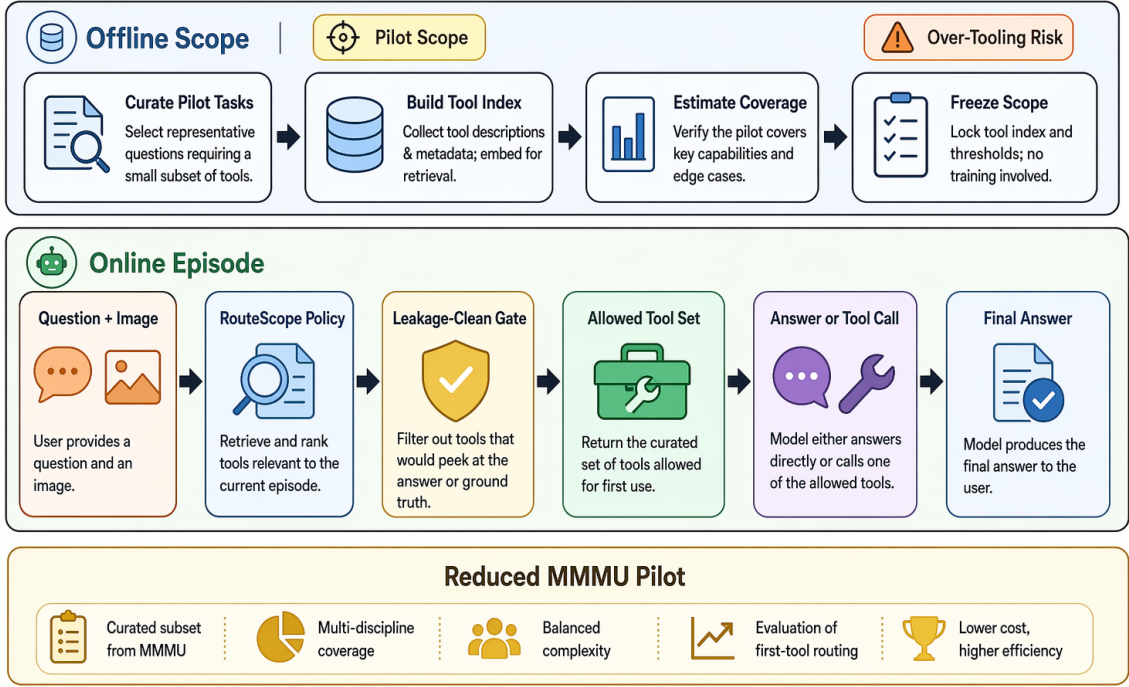


Figure 1: RouteScope first-action routing overview. The evaluated policy chooses between direct answering and tool-mediated evidence before producing a benchmark-scored answer, while the row contract blocks benchmark metadata, gold answers, and evaluator feedback from routing inputs.

back, or any field created after scoring. This contract prevents the router from becoming a dataset recognizer. It also makes negative results more informative: if RouteScope underperforms, the failure is attributable to the deployable routing signal and answer path rather than to hidden metadata leakage.

Each tool branch has a different failure mode. OCR can help when the answer depends on visible text, but it can also compress layout and lose non-text visual evidence. VQA can focus the model on a visual question, but it may introduce an intermediate answer that is too narrow for the final prompt. Zoom can recover local detail, but it can discard global context. Captioning and set-of-mark style controls can expose object structure, but may smooth away quantitative details. These branch-specific tradeoffs motivate reporting family-level rankings instead of a single method order.

4 Experimental Setup

The revised matrix evaluates three executed benchmark families: MMMU/MMMU-Pro, OCR-Bench/OCRBench variants, and MathVista. Each family contributes 20 selected tasks, yielding 60

task instances. Eight method conditions are evaluated on each task, giving 480 expected method-task rows. The metric is task-level accuracy after family-specific answer normalization or scoring. All conditions use the same hosted multimodal response backend, with one planned model call per method-task row, a 4096-token output cap, and no explicit temperature override. These settings make the comparison a controlled first-action study rather than a comparison between different response backends.

The matrix includes Direct Answer, Always OCR, Always VQA, Always Zoom, MM-REACT Style, ViperGPT Style, VWA SoM Caption, and RouteScope. We report both aggregate and family-specific means because the selected families stress different tool affordances. OCRBench naturally emphasizes text extraction, MMMU combines subject knowledge with visual reasoning, and MathVista emphasizes mathematical visual reasoning. VisualWebArena is not part of the executed result set. Its exclusion is a scoped family revision, not a negative score.

One Always-OCR row on a MMMU-Pro finance task has no scored answer because the response ended at the configured output cap. We count the

row in expected rows and null rows, exclude it from score means, and do not fill a score manually.

Table 1 summarizes the executed sources. The most important scope detail is the family mix. VisualWebArena remains a target benchmark for multimodal web agents, but it is not part of the reported result set. Treating that absence as a benchmark score would be misleading. The paper instead reports the executable MathVista replacement and constrains every result sentence to the selected MMMU/OCRBench/MathVista family mix.

The scoring policy is similarly conservative. Missing answers are not manually filled. Result summaries report expected rows, scored rows, null rows, and mean scores separately. This is why the Results section can state 480 expected rows and 479 scored rows at the same time: the missing score remains part of row accounting even though it is excluded from mean accuracy.

The selected families also differ in what a first tool can plausibly recover. MMMU rows often combine image evidence with domain concepts and answer options, so a visual summary can help only when it preserves the relevant concept. OCRBench rows reward text extraction but can still fail when the extracted string is incomplete or when the answerer overfits noisy text. MathVista rows frequently require spatial or mathematical interpretation, making zoom useful on some examples while direct answering remains strong on others. This heterogeneity is why the paper reports both the aggregate matrix and the per-family matrix, rather than only a single headline number.

The scoring protocol is intentionally simple. Each method receives the same task row and budget policy, produces one final answer, and is scored by the benchmark-facing answer normalizer or exact-match scorer for that family. We do not use cross-row adaptation, per-family oracle selection, or post-hoc method switching. This keeps the result interpretable as a first-action utility audit: if a fixed tool or RouteScope changes accuracy, the change comes from the evidence exposed to the answerer under that method condition.

5 Results

Table 2 is the binding paper-facing result matrix. It reports the current selected-family evidence, not an earlier pilot and not VisualWebArena execution. The aggregate leader is VWA SoM Caption at 41.7% accuracy. RouteScope scores 31.7%, be-

low Direct Answer at 33.3% by 1.7 absolute points and below the aggregate leader by 10.0 absolute points. The aggregate therefore supports a negative, diagnostic interpretation rather than a RouteScope improvement claim.

The family-specific view explains why the aggregate should not be narrated as a uniform routing win. Always Zoom leads MathVista at 65.0%, VWA SoM Caption leads MMMU at 55.0%, and several OCRBench conditions tie near 10.0% while Direct Answer and ViperGPT Style are at 5.0%. RouteScope is competitive on MMMU and OCRBench relative to Direct Answer, but it trails Direct Answer on MathVista and trails the strongest aggregate control overall.

The MathVista slice is the clearest warning against a generic “route to tools” conclusion. Direct Answer reaches 60.0%, VWA SoM Caption also reaches 60.0%, and Always Zoom reaches 65.0%, but RouteScope reaches 45.0%. The router therefore fails to select the high-utility branch often enough on this family. This does not mean tools are useless; it means the evaluated routing signal is weaker than the best fixed visual transformation on MathVista.

The MMMU slice points in the opposite direction but is not enough to rescue the aggregate. RouteScope reaches 40.0%, above Direct Answer at 35.0%, yet below VWA SoM Caption at 55.0% and Always Zoom at 45.0%. The correct reading is that first-tool choice matters on MMMU, but RouteScope is not the best available policy among the evaluated controls. This is exactly the kind of family-specific utility the matrix was designed to expose.

OCRBench remains a low-accuracy slice for all methods in this selected subset. Always VQA, Always Zoom, MM-REACT Style, VWA SoM Caption, and RouteScope each reach 10.0%, while Direct Answer, Always OCR, and ViperGPT Style reach 5.0%. The low absolute values prevent strong claims about OCR-specific routing, but the tie pattern still matters: RouteScope does not underperform Direct Answer on this family, yet it also does not separate itself from simple fixed-tool controls.

Control-specific interpretation. The results are easier to read as a utility map than as a method leaderboard. Direct Answer is the required lower-complexity control because it preserves the original visual context and exposes no tool-generated evidence before answering. RouteScope must beat

Family	Source and scoring role	Tasks	Model/budget	Metric	Scope note
MathVista	Testmini visual-math rows with normalized answer matching	20	Shared hosted back-end; 4096-token cap	Accuracy	Replacement family for the excluded VisualWebArena execution slot; not web-interaction evidence.
MMMU	MMMU and MMMU-Pro selected rows with dataset answer fields	20	Shared hosted back-end; 4096-token cap	Accuracy	Expert multimodal reasoning rows; includes the residual Always-OCR null in expected-row accounting.
OCRBench	OCRBench and OCRBench-v2 selected rows with official answer fields	20	Shared hosted back-end; 4096-token cap	Accuracy	Text-heavy visual QA slice; all method rows scored after collection.

Table 1: The executed matrix covers three public benchmark families and 480 expected method-task rows under the same hosted backend and output budget; VisualWebArena is excluded, and MathVista is a scoped replacement rather than equivalent web evidence.

Method	MathVista	MMMU	OCRBench	Aggregate
Direct Answer	60.0	35.0	5.0	33.3
Always OCR	55.0	31.6	5.0	30.5
Always VQA	50.0	40.0	10.0	33.3
Always Zoom	65.0	45.0	10.0	40.0
MM-REACT Style	35.0	30.0	10.0	25.0
ViperGPT Style	25.0	35.0	5.0	21.7
VWA SoM Caption	60.0	55.0	10.0	41.7
RouteScope	45.0	40.0	10.0	31.7

Table 2: Across 480 expected rows under one shared hosted backend and a 4096-token output cap, 479 are scored; RouteScope reaches 31.7% aggregate accuracy, below Direct Answer at 33.3%, while VWA SoM Caption leads at 41.7% on the selected families. Always OCR has one MMMU null, so its MMMU and aggregate means are computed over 19 and 59 scored rows respectively.

this condition to justify routing overhead, and in the current aggregate it does not. The gap is small in absolute terms, but the sign matters: the evaluated router loses one more paired task to Direct Answer than it wins, and the aggregate mean is 1.7 points lower.

Always Zoom and VWA SoM Caption show why the negative RouteScope result is not a negative result for tools in general. Always Zoom is second overall and best on MathVista, which suggests that local visual detail can be useful for the selected visual-math rows. VWA SoM Caption is the strongest aggregate condition and best on MMMU, which suggests that structured visual summaries can preserve helpful object and layout evidence on expert multimodal questions. These controls are fixed policies rather than adaptive routers, so their success makes the routing problem harder: a first-action policy must learn when to choose them and when to stay direct.

The weaker literature-style controls also matter. MM-REACT Style and ViperGPT Style are included as controlled method conditions inspired by multimodal tool-use systems, not as exact reproductions of the original papers. Their lower aggregate scores show that a more elaborate tool-use pattern is not automatically better under a one-answer scoring protocol. This is a useful warning for tool-orchestration papers: invoking a tool taxonomy can make a trajectory look more agentic while still reducing answer accuracy.

Family-specific interpretation. MathVista drives much of the aggregate loss for RouteScope. The family contains useful transformations, since Always Zoom reaches 65.0% and Direct Answer and VWA SoM Caption each reach 60.0%, but RouteScope reaches only 45.0%. The error is therefore not simply that MathVista is hostile to tools. It is that the router does not select or exploit

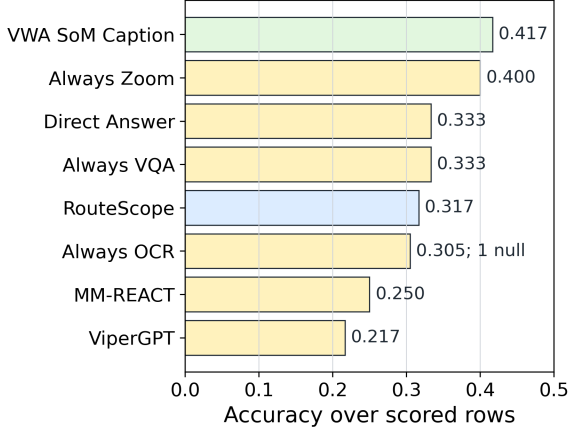


Figure 2: Aggregate method accuracy from the selected-family analysis. The plot visualizes the same null-excluded means as Table 2 and highlights that RouteScope is not the aggregate leader.

the stronger branch often enough on this slice.

MMMU gives the most favorable evidence for RouteScope, but only at a narrow scope. RouteScope reaches 40.0%, above Direct Answer at 35.0%, while still below VWA SoM Caption at 55.0% and Always Zoom at 45.0%. The correct conclusion is that first-tool choice can help selected expert multimodal rows, not that the current router is the best first-tool policy. OCRBench is flatter: several methods tie at 10.0%, and the small spread prevents a confident OCR-specific story. Together, the three families support a paper about heterogeneous tool utility rather than a paper about consistent routing improvement.

6 Analysis

Paired binary tests reinforce the conservative interpretation. Against Direct Answer over all selected families, RouteScope has five tasks where only RouteScope is correct and six where only Direct Answer is correct, with exact McNemar-style binomial p-value 1.000. Against VWA SoM Caption, RouteScope has zero RouteScope-only wins and six caption-control-only wins, with p-value 0.031. These tests are single-run diagnostics with the residual null excluded, so they support claim calibration rather than broad statistical generalization.

The failure taxonomy is small but important. Five MMMU rows initially lacked scores because responses ended at the output cap; four became scored before the final summary, while one Always-OCR row on a MMMU-Pro finance task remained null. Because that row belongs to a baseline condition, the aggregate comparison is not rescued

or harmed for RouteScope. The paper therefore keeps the matrix scoped as null-excluded rather than clean.

This residual-null treatment affects the strength of the paper’s claims. The Always-OCR aggregate is computed over 59 scored rows rather than 60, so it is not directly described as a clean full-row condition. The other seven methods have 60 scored rows each. When comparing RouteScope with Direct Answer or VWA SoM Caption, the paired diagnostics use tasks where both relevant methods are scored and keep the residual null outside the mean. This is conservative: it neither invents a score for the missing baseline row nor uses the missing row to improve RouteScope’s position.

The current evidence also clarifies which next experiment has the highest value. A RouteScope-vs-context ablation should distinguish two possible causes of underperformance: the router may choose the wrong branch, or the final answerer may fail to use the routed evidence once selected. A residual-null cap diagnostic should test whether the remaining Always-OCR failure is an output-budget artifact. A VisualWebArena-inclusive evaluation should wait until the benchmark environment is available, because otherwise it would mix benchmark-family evidence with environment failure.

This analysis changes the paper story. A positive RouteScope paper would argue that lightweight routing reliably selects the right first tool. The current matrix instead shows that simple controls are hard to beat: a captioning-style visual control is best on the aggregate, zoom is best on MathVista, and Always VQA ties or leads the OCRBench slice. RouteScope’s value in the current evidence is therefore methodological. It provides a controlled row contract for asking where a first tool helps, but the evaluated policy is not yet the answer to that question.

The result also helps prioritize repairs. Adding more prose cannot turn the current matrix into a win. The next evidence-bearing improvements are targeted experiments: a context ablation to locate the failure point, a VisualWebArena readiness repair to restore the web-agent family, and an output-cap diagnostic for the single remaining null. Until those are run, the appropriate claim is that RouteScope exposes measurable first-tool failure modes under the current row contract.

Comparison	Family	Pairs	RS only	Other only	p
RouteScope vs Direct Answer	All selected	60	5	6	1.000000
RouteScope vs VWA SoM Caption	All selected	60	0	6	0.031250
RouteScope vs Direct Answer	MathVista	20	1	4	0.375000
RouteScope vs VWA SoM Caption	MathVista	20	0	3	0.250000
RouteScope vs Direct Answer	MMMU	20	3	2	1.000000
RouteScope vs VWA SoM Caption	MMMU	20	0	3	0.250000
RouteScope vs Direct Answer	OCRBench	20	1	0	1.000000
RouteScope vs VWA SoM Caption	OCRBench	20	0	0	1.000000

Table 3: Paired diagnostics show no aggregate RouteScope advantage over Direct Answer: five RouteScope-only wins versus six Direct-only wins over 60 paired tasks.

What the paired tests rule out. The paired diagnostics are not large enough to support broad benchmark generalization, but they do rule out the easiest positive narrative for this evaluation. If RouteScope were merely hurt by a few unlucky family averages, it should still show a favorable paired pattern against Direct Answer. It does not: over 60 paired tasks, there are five RouteScope-only wins and six Direct-only wins. The comparison with VWA SoM Caption is sharper, with no RouteScope-only wins and six caption-only wins. This means the aggregate ordering is not only a consequence of averaging across families; it is also visible at the paired-task level for the strongest caption-style control.

The residual null does not explain away this pattern. The missing score belongs to Always OCR, not to RouteScope, Direct Answer, or VWA SoM Caption. RouteScope has 60 scored rows, Direct Answer has 60 scored rows, and VWA SoM Caption has 60 scored rows. The clean paired comparisons among those three conditions therefore remain intact. The null affects how we describe Always OCR and the overall row accounting, but it is not a hidden advantage or penalty for the central RouteScope comparison.

The family-level paired counts also show why a single aggregate sentence is insufficient. On MathVista, Direct Answer has four task-only wins over RouteScope while RouteScope has one, matching the family mean gap and explaining why zoom and direct visual context remain hard to beat. On MMMU, RouteScope has three task-only wins over Direct Answer and two losses, so the router has some local value, but the caption-style control still has three caption-only wins and no RouteScope-only wins. OCRBench is flatter: RouteScope has one task-only win over Direct Answer and no losses, yet all leading methods stay near 10.0%

accuracy. These counts support the same diagnosis as the family means: first-tool choice matters, but the current router does not reliably pick the strongest evidence source across families.

7 Failure Cases and Threats to Validity

The clearest failure case is not an incorrect answer but a missing score. Always OCR on one MMMU-Pro finance row exhausted the configured output budget and returned no text to score. The result is not hidden: it is counted as one expected row, one null row, and zero contribution to the mean. This treatment is stricter than converting the row to an incorrect answer because the underlying failure is a generation/return failure rather than an answer that the benchmark scorer judged wrong. It is also stricter than dropping the row silently because the matrix continues to report 480 expected rows.

A second threat is that the method labels are not equivalent in semantic scope. VWA SoM Caption is a method condition inspired by set-of-mark/captioning style visual evidence. It is not a VisualWebArena benchmark result. Without this distinction, a reader could incorrectly infer that the matrix contains web-interaction evidence. The paper repeats the distinction because the aggregate leader uses a VWA-named method while the VisualWebArena family is excluded.

A third threat is task selection. The current matrix uses 20 selected tasks per family, which is enough to reveal large rank shifts but not enough for broad benchmark claims. This is why the paired tests are described as diagnostics and why the conclusion avoids general statements about all multimodal agents. A final submission version should either increase task coverage or keep the claim narrow: the evaluated training-free policy underperforms Direct Answer on this selected family mix.

The selected-family mix also changes what “tool use” means across rows. MMMU rows often depend on answer choices and domain concepts, MathVista rows can depend on local geometry or symbolic reasoning, and OCRBench rows reward text extraction. A single router can therefore fail in several different ways. It may choose a tool that is globally plausible but locally unnecessary; it may choose a tool that preserves the wrong part of the image; or it may expose useful evidence that the final answerer fails to use. The present matrix can identify the resulting accuracy pattern, but it cannot by itself attribute each error to routing, perception, or answer synthesis.

The backend and budget policy are another comparability boundary. All reported rows use the same hosted multimodal response backend, one planned model call per method-task row, and the same 4096-token output cap. This shared backend makes the method conditions comparable within the run, but it does not claim that the same rank order would hold for a different base model, a multi-call planner, a different output budget, or a tool executor with stronger intermediate verification. The negative result should therefore be read as a property of this controlled first-action setting, not as a theorem about all routing policies.

The VWA SoM Caption condition requires especially careful naming. It is the aggregate leader in this matrix, but it is a method condition, not a VisualWebArena family result. The condition exposes structured visual/caption evidence on the selected MMMU, OCRBench, and MathVista rows. Because the VisualWebArena family is absent, this result cannot be used to claim web-task competence, browser-state recovery, or realistic web-agent success. Its scientific value is narrower: it shows that a caption-style first evidence source is a strong non-adaptive control on the executed families.

The residual null creates a different kind of limitation. Since the missing row belongs to Always OCR, it does not affect the clean RouteScope–Direct Answer or RouteScope–VWA SoM Caption paired comparisons. It does affect the claim that the matrix is fully scored. For that reason, the paper reports both expected rows and scored rows and avoids describing Always OCR as a clean 60-row condition. This distinction is small numerically but important procedurally: a generation failure should not be silently converted into either a wrong answer or a missing denominator.

Finally, the current method does not ex-

pose a learned or calibrated confidence model. RouteScope chooses a first branch from task-facing signals, but the matrix does not report uncertainty over that choice or a confidence threshold for falling back to Direct Answer. A future version could use calibration without changing the training-free premise: for example, the router could abstain when question and visual cues do not clearly favor a tool, or it could require agreement between direct and tool-mediated evidence before committing. The present result motivates such mechanisms because fixed controls outperform the adaptive router on the aggregate.

Finally, the matrix does not yet isolate whether RouteScope fails at branch selection or answer use. If the router chooses a useful branch but the answerer ignores the resulting evidence, a different answer-context design could recover accuracy without changing routing. If the router chooses the wrong branch, the policy itself needs better features or calibration. The deferred RouteScope-vs-context ablation is therefore not cosmetic; it is necessary for turning the negative result into a causal diagnosis.

8 Conclusion

RouteScope tests a narrow but important question for multimodal agents: whether a training-free first-action policy can choose the evidence source that improves final answer accuracy. In the revised 480-row matrix over MMMU, OCRBench, and MathVista, the answer is currently no. RouteScope reaches 31.7% aggregate accuracy, below Direct Answer at 33.3%, while fixed zoom and caption-style controls expose stronger family-specific behavior. This negative result is still useful because it identifies where first-tool orchestration fails and which controls remain hard to beat. The next step is not to relabel the result as a win, but to isolate route selection from answer-context use, evaluate VisualWebArena under its own benchmark protocol, and resolve the remaining output-cap null through normal scoring.

9 Limitations

The matrix is not a clean final result because one reported residual null remains after replay. VisualWebArena is excluded, so no claim in this paper should be read as visual-web execution evidence. The current method set also lacks some stronger frontier ablations and literature baselines needed

for causal claims about why the router underperforms or which context feature causes a routing decision.

Ethical Considerations

Tool-routing agents can amplify benchmark leakage, web automation risk, and tool-output errors if their inputs or actions are not constrained. The evaluated row contract blocks gold answers, evaluator outputs, source identifiers, and benchmark-family labels from routing inputs. Web-interaction evaluation should resume only when an official reset and scoring protocol are available, so that benchmark state is not polluted by uncontrolled automation.

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A Detailed Family Results

Table 4 expands the body result matrix into one row per family and method. It is included to make the rank-shift claim auditable without putting a dense 24-row table in the main body.

Table 4: Family-method matrix for the selected full run; RouteScope is rank 5 on aggregate and does not lead any executed family.

Family	Method	Acc.	Scored	Family rank	Agg. rank
MathVista	Always Zoom	65.0	20/20	1	2
MathVista	Direct Answer	60.0	20/20	2	4
MathVista	VWA SoM Caption	60.0	20/20	3	1
MathVista	Always OCR	55.0	20/20	4	6
MathVista	Always VQA	50.0	20/20	5	3
MathVista	RouteScope	45.0	20/20	6	5
MathVista	MM-REACT Style	35.0	20/20	7	7
MathVista	ViperGPT Style	25.0	20/20	8	8
MMMU	VWA SoM Caption	55.0	20/20	1	1
MMMU	Always Zoom	45.0	20/20	2	2
MMMU	Always VQA	40.0	20/20	3	3
MMMU	RouteScope	40.0	20/20	4	5
MMMU	Direct Answer	35.0	20/20	5	4
MMMU	ViperGPT Style	35.0	20/20	6	8
MMMU	Always OCR	31.6	19/20	7	6
MMMU	MM-REACT Style	30.0	20/20	8	7
OCRBench	Always VQA	10.0	20/20	1	3
OCRBench	Always Zoom	10.0	20/20	2	2
OCRBench	MM-REACT Style	10.0	20/20	3	7
OCRBench	RouteScope	10.0	20/20	4	5
OCRBench	VWA SoM Caption	10.0	20/20	5	1
OCRBench	Always OCR	5.0	20/20	6	6
OCRBench	Direct Answer	5.0	20/20	7	4
OCRBench	ViperGPT Style	5.0	20/20	8	8

B Paired Diagnostic Details

Table 5 lists every paired diagnostic generated for RouteScope comparisons. These rows are single-run binary paired diagnostics, not a broad significance claim.

C Failure and Scope Notes

The residual null is Always OCR on one MMMU-Pro finance task. The response ended at the configured output cap and produced no answer to score, so the row is counted in expected-row and null-row accounting but excluded from mean accuracy. Four other initially missing MMMU scores became scored before the final summary. This note is included so the reader can distinguish a scoped warning from a hidden scoring edit.

The family-scope note is equally important. VisualWebArena is excluded from the reported evidence. MathVista is the executable replacement family for this matrix; it is not equivalent web-interaction evidence. The manuscript therefore never reports VisualWebArena task success and never treats a VWA-named method condition as VisualWebArena family execution.

Family	Comparison	Pairs	RouteScope only	Other only	p
All selected	RouteScope vs Direct Answer	60	5	6	1.000
All selected	RouteScope vs Always OCR	59	4	3	1.000
All selected	RouteScope vs VWA SoM Caption	60	0	6	0.031
MathVista	RouteScope vs Direct Answer	20	1	4	0.375
MathVista	RouteScope vs Always OCR	20	0	2	0.500
MathVista	RouteScope vs VWA SoM Caption	20	0	3	0.250
MMMU	RouteScope vs Direct Answer	20	3	2	1.000
MMMU	RouteScope vs Always OCR	19	3	1	0.625
MMMU	RouteScope vs VWA SoM Caption	20	0	3	0.250
OCRBench	RouteScope vs Direct Answer	20	1	0	1.000
OCRBench	RouteScope vs Always OCR	20	1	0	1.000
OCRBench	RouteScope vs VWA SoM Caption	20	0	0	1.000

Table 5: Paired diagnostics with the residual null excluded from affected comparisons.

D Method-Condition Notes

Direct Answer is the minimal control: it answers from the original task context without first exposing a tool-mediated representation. Its role is not to be a weak baseline but to test whether routing changes the evidence enough to justify any additional transformation.

Always OCR, Always VQA, and Always Zoom are fixed first-action controls. They are deliberately simple because they make family-level affordances visible. If a fixed control beats RouteScope, the correct diagnosis is not that tools are useless; it is that the adaptive selector did not reliably choose or use a useful branch under the current row contract.

MM-REACT Style and ViperGPT Style are literature-inspired controls included to test whether richer tool-use patterns help in a one-answer scoring protocol. They are not exact reproductions of the original systems, and their lower aggregate scores should not be read as a claim about those systems. They show only that adding a tool-use pattern without calibrated first-action selection can reduce answer accuracy in this setting.

VWA SoM Caption is a captioning/set-of-mark style method condition. The label is retained because the condition was designed as a structured visual evidence control, but the reported rows are still MMMU, OCRBench, and MathVista rows. This distinction is central to the current scope: the method condition can be strong on the selected families while the VisualWebArena benchmark family remains absent.

E Benchmark-Family Notes

MMMU and MMMU-Pro rows stress expert multimodal reasoning with answer choices and domain concepts. The selected MMMU slice is where RouteScope most clearly beats Direct Answer, but the stronger caption-style and zoom controls show that the router is not yet selecting the best available first evidence source.

OCRBench and OCRBench-v2 rows stress text-heavy visual QA. The low absolute scores in the selected OCRBench slice make the family useful as a warning rather than a positive result. RouteScope ties several fixed controls at 10.0%, but the whole slice remains too low to support a strong OCR-routing claim.

MathVista rows stress visual-mathematical reasoning. This family produces the clearest RouteScope failure: Direct Answer and caption-style evidence each reach 60.0%, Always Zoom reaches 65.0%, and RouteScope reaches 45.0%. The family therefore motivates a branch-selection diagnosis rather than a generic statement that tool use helps.

VisualWebArena remains outside the executed matrix. A VisualWebArena-inclusive study should be reported as a separate benchmark-family evaluation, not inferred from the current VWA-named method condition.

F Claim Calibration Notes

The paper keeps three claims and rejects three tempting ones. It keeps the terminal selected-family evidence claim, the family-specific ranking claim, and the claim that RouteScope underperforms Direct Answer on aggregate. It rejects the claims that RouteScope is an aggregate winner, that the matrix is null-free, and that the current evidence contains VisualWebArena execution.

The backend and budget boundary is separate from the arithmetic of the selected-family tables. All method rows share one hosted backend and output cap, so the within-run comparisons are controlled, but the reported rank order should not be generalized across base models or serving policies.

The residual-null gap is also kept narrow. The remaining null belongs to Always OCR, so it does not alter the clean paired RouteScope–Direct Answer or RouteScope–caption comparisons. It does, however, prevent any claim that all 480 expected rows were scored. The paper therefore reports 479 scored rows and one audited null in every summary artifact.

The deferred RouteScope-vs-context ablation is the most important scientific follow-up. If RouteScope chose reasonable branches but the answerer failed to use routed evidence, answer-context design is the bottleneck. If RouteScope chose weak branches, the routing policy itself needs better cues or calibration. The current study stops before that causal claim.